

MINISTRY OF DEFENCE AND NATIONAL SERVICE
MILITARY SCHOOLS ASSOCIATION
FORM TWO PRE-NATIONAL EXAMINATION 2026
CHEMISTRY - MARKING SCHEME

SECTION A: (15 MARKS)

1.

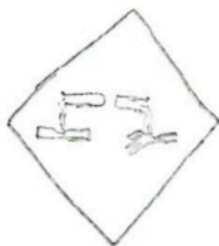
i	ii	iii	iv	v	vi	vii	viii	ix	x	0.10
C	A	D	B	A	A	AB	D	A	C	

2.

i	ii	iii	iv	v	0.20
E	H	G	B	C	

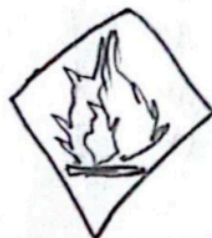
3. a. (i) MN_2 0.1 mark
 (ii) electrovalent/ionic bonds 0.1 mark
 (iii) - It conducts electricity
 0.1 mark - It is formed between metal and non-metal
 - It has high boiling and melting point

b. (i)



0.1 mark - 1.5 mark
CORROSIVE

(ii)



Flammable

0.1 mark

1.5 mark

(ii)



TOXIC

0.1 mark

1.5 mark

4. a. (i) Electron 2 mark

- (ii) - Detect diseases early to fractures
 - Diagnose injuries to internal bleeding
 - Guide treatment to help to plan surgery.
- Any 2 points - 2 mark

b. (i) - Sublimation

(ii) - Filtration

(iii) - Solvent extraction

0.5 mark @

(iv) - Fractional distillation

(v) - Paper Chromatography

(vi) - Evaporation

5. a. (i) - Ability of an atom to attract electrons towards itself. 0.1 mark

(ii) - Is minimum amount of energy required to remove the most loosely electron from a gaseous atom 1.5 mark

b.

I	II	III	IV	V	VI	VII	VIII
A						B	C
			D				
E	F						

0.5 mark @

(i) - F

(ii) - C

(iii) - A

0.5 mark @

(iv) - B

(v) - F

(vi) - D

6. a. (i) Group 3 and period 3 01 mark
 (ii) - 15 electrons 01 mark
 (iii) - 16 neutrons 01 mark
 (iv) Is a non-metal because it is found in group 3 and does not conduct electricity 01 mark
 (v) ${}_{15}^{31}\text{X}$ 02 mark

b. Data given

$$P_1 = 90.5\%, P_2 = 0.3\%, P_3 = 9.2\%$$

$$\text{R.A.M} = (P_1 \times A_1) + (P_2 \times A_2) + (P_3 \times A_3) \quad 01 \text{ mark}$$

$$= \left(\frac{90.5 \times 20}{100} \right) + \left(\frac{0.3 \times 21}{100} \right) + \left(\frac{9.2 \times 22}{100} \right) \quad 01 \text{ mark}$$

$$18.1 + 0.063 + 2.024$$

$$\text{R.A.M} = 20.2 \quad 01 \text{ mark}$$

- 07 a. - Oxidation is the loss of electrons - 0.5 mark
 - Reduction is the gain of electrons - 0.5 mark

- b. (i) They show the correct number of atoms involved
 (ii) They show the chemical formulas of reactants and products 01 mark @
 (iii) They can indicate the physical states of substances
 (iv) They allow you to calculate the amounts of substances in a reaction.

c. (i) Decomposition reaction

(ii) Combination reaction

(iii) Displacement reaction

(iv) Precipitation reaction

01 mark @

10. a. (i) Is the formula which represents the simplest ratio of the atoms or ions in a compound. 01 mark

(ii) Is the formula that shows the actual number of atoms in a molecule. It is a multiple of empirical formula. 01 mark

b. (i) - Na_2O

(ii) - $\text{Ca}(\text{NO}_3)_2$ 01 mark @

(iii) - NH_4Cl

(iv) - CCl_4

c. Data given

Mass of N = 30.4g

Mass of O = 69.6g

R.A.M N = 14

R.A.M O = 16

Relative density = 46%

from
30.4g
14

2.17
2.1

1

1

69.6g
16

4.35
2.1

2.07

2

01 mark

(i) N = 1, O = 2. $\Rightarrow \text{NO}_2$ 01 mark
Empirical formula is NO_2 .

(ii) Molecular formula.

$$\text{MF} = \text{Empirical formula} \times n$$

10 c. But $R.M.M = n \times (\text{Sum of R.A.M.})$.

where $R.M.M = \text{vapour density} \times 2$. 01 mark
 $= 46 \times 2 = 92$.

$$92 = n(\text{NO}_2).$$

$$92 = n(14 + (16)_2)$$
$$= n(14 + 32) \quad 01 \text{ mark}$$

$$92 = 46n$$
$$\frac{92}{46} = \frac{46n}{46}$$

$$n = 2.$$

$$M.F = \sum F \times n.$$

$$= (\text{NO}_2)_2$$

$$= \text{N}_2\text{O}_4. \quad 01 \text{ mark}$$

(ii) Molecular formula is N_2O_4 .

(iii) The compound name is
Dinitrogen tetroxide. 01 mark